Case 5 — Mr. Leonard Vance

66 years old · Male | Parkinson's Disease · Diagnosed 12 months ago

Target: 70%+ | Neurological Physiotherapy

■ PROBLEM LIST

- Resting tremor — right hand (4–6 Hz)
- Bradykinesia — slowed, reduced-amplitude movement
- Akinesia — difficulty initiating movement (chair)
- Reduced right arm swing during gait
- Stooped posture — thoracic kyphosis, flexor dominance
- Decreased bilateral hip and knee flexor ROM
- Micrographia — fine motor hypokinesia
- Likely cogwheel/leadpipe rigidity (to confirm objectively)
- Reduced walking speed — cannot keep pace with wife
- Difficulty initiating sit-to-stand from low chairs
- Impaired handwriting — impacts ADLs (correspondence, forms)
- Reduced participation in social/community walking

■ RED FLAGS

None identified

■ PATHOPHYSIOLOGY

• 2 Pathophysiology of Working Diagnosis What Is Happening Neurologically Parkinson's Disease is a progressive neurodegenerative disorder characterised by the selective loss of dopaminergic neurons in the substantia nigra pars compacta (SNc).

• This thalamic suppression dampens motor cortex excitation, producing the cardinal features of hypokinesia, bradykinesia, and akinesia.

• Vance's diagnosis, the disease process was likely well established for years before his resting tremor became the presenting complaint.

• Typical Signs and Symptoms — Explained by Pathophysiology Resting tremor (4–6 Hz, "pill-rolling"): Arises from pathological oscillations in the thalamo-cortical and basal ganglia-thalamic circuits when normal dampening by the dopaminergic system is lost.

• Suppressed during voluntary movement because cortical volitional drive temporarily overrides the oscillation — this is why the tremor disappears when he reaches for a cup.

■ DIFFERENTIAL DIAGNOSES

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■ OBJECTIVE ASSESSMENTS & HOW TO PERFORM

Timed Up and Go (TUG)

Test 1: Timed Up and Go (TUG) What it measures: A composite measure of functional mobility including sit-to-stand, walking, turning (180°), walking back, and returning to sitting. It captures gait initiation speed, walking velocity, turning ability, and sit-to-stand — exactly the

Pull Test (Postural Instability — MDS-UPDRS Item 3.12)

Test 2: Pull Test (Postural Instability — MDS-UPDRS Item 3.12) What it measures: Integrity of automatic postural righting reflexes — the ability to recover from an unexpected backward perturbation without taking excessive corrective steps or falling. Directly assesses a key domai

Finger-Tapping Speed & Amplitude (MDS-UPDRS Item 3.4)

Test 3: Finger-Tapping Speed & Amplitude (MDS-UPDRS Item 3.4) What it measures: Bradykinesia of the right upper limb — specifically the speed, amplitude, and pattern of repetitive sequential finger movements. A decremental pattern (progressively slowing and decreasing amplitude o

■ TREATMENT PLAN

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